

Grade 9 Physical Science - Forces and Motion - Short Test

Time: 25 minutes. Total: 10 marks. Mark one letter for each multiple-choice question. Write your written answers in the spaces on the answer sheet.

Q1. An object is moving at a steady speed in a straight line. The net force on it is zero.

What will the object do next?

- A. Stop immediately
- B. Slow down gradually
- C. Keep moving at the same steady speed
- D. Speed up

Q2. Which unit is used to measure force?

- A. joule
- B. newton
- C. watt
- D. pascal

Q3. A 2 kg trolley accelerates at 3 m/s squared. What is the net force on it?

- A. 1.5 N
- B. 5 N
- C. 6 N
- D. 9 N

Q4. A book rests on a table. Using forces, explain why the book is not accelerating. [3 marks]

Q5. A cyclist rides at a steady speed along a flat, straight road.

(a) The forward force from the pedals is ... the total resistive force. Mark one letter.

- A. greater than
- B. equal to
- C. less than

(b) Explain your choice in (a). [3 marks]